

# Queues in coffee shops



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## Abstract

The effect the way the tasks are divided up between the servers in a small coffee shop on the time it takes to deal with the needs of customers are considered. We find that it takes longer to serve the needs of customers when one server is responsible for taking customer orders, one is responsible for making teas and one is responsible for making coffees. We therefore conclude that the cafe is more efficient if the three servers who work in the cafe take customer orders and then make the drinks that customers require.

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## 1 Introduction

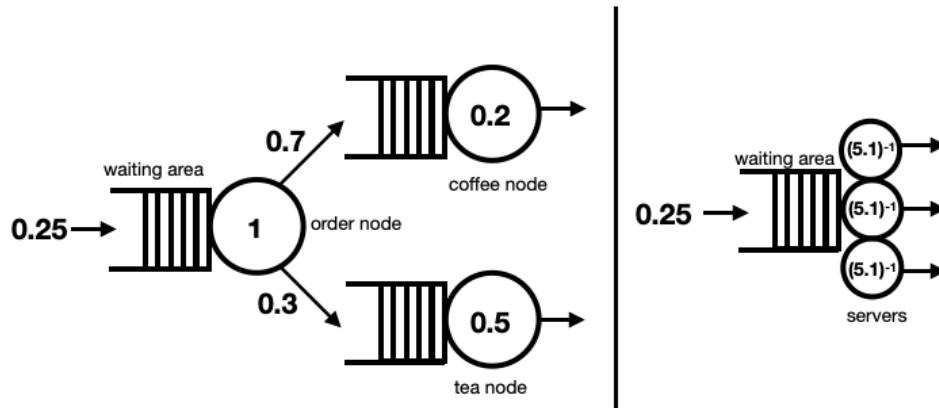
Customers do not like waiting in long lines to receive services. Furthermore, as humans do not need to drink coffee or tea or order to survive, a customer who enters a coffee shop and sees a long queue is highly likely to leave. Thus, coffee shops have a clear incentive to minimise the length of the lines in their stores.

In this report, I have used data about customer behaviour and event-based simulation to construct models for investigating how long customers will have to queue to get a drink. I use this model to examine whether it is better for the staff to take on specialised roles in the shop or not. The report is laid out as follows: in section 2, I explain the two ways of organising the servers that I have considered and the models for the customer waiting time that I have constructed based on these different ways of distributing tasks in the cafe. By simulating the models, I show, in section 3, that it is better if the three staff members do not take on specialised roles.

## 2 Models

This report considers two ways of dividing the tasks in the shop. The first way has each server specialise in performing only one task. In other words, one server is responsible for taking the orders, one server is responsible for making cups of tea, and the final server is responsible for making cups of coffee. The left panel of figure 1 illustrates the system of queues that we have to model to understand the queuing time in this model.

In constructing this model, we have assumed that customers arrive according to a Poisson process with a rate of  $0.25 \text{ minutes}^{-1}$ . This number was obtained by taking the average rate of



**Fig. 1.** The two ways of operating the coffee shop that are considered in this report. The left panel shows the model in which one staff member takes customers' orders, one makes tea, and the other makes coffee. The right panel illustrates the model with no specialisation. In this model, each server takes on all three tasks. The numbers in the figures indicate the arrival and service rates in units of inverse minutes or the transition probabilities, which are unit less.

arrivals per hour in a typical day. On arrival, they join a queue and wait to give their order to a server. The time it takes for a customer to give their order,  $S$ , is modelled as an exponential random variable with  $\mathbb{E}(S) = 1$  minute. Having given their order, 70% of customers join the queue to get a coffee while the remaining customers join the line to get a cup of tea. The amount of time it takes to make a customer's coffee,  $C$ , is an exponential random variable with  $\mathbb{E}(C) = 5$  minutes. In contrast, the amount of time it takes to make a cup of tea,  $T$  is an exponential random variable with  $\mathbb{E}(T) = 2$  minutes.

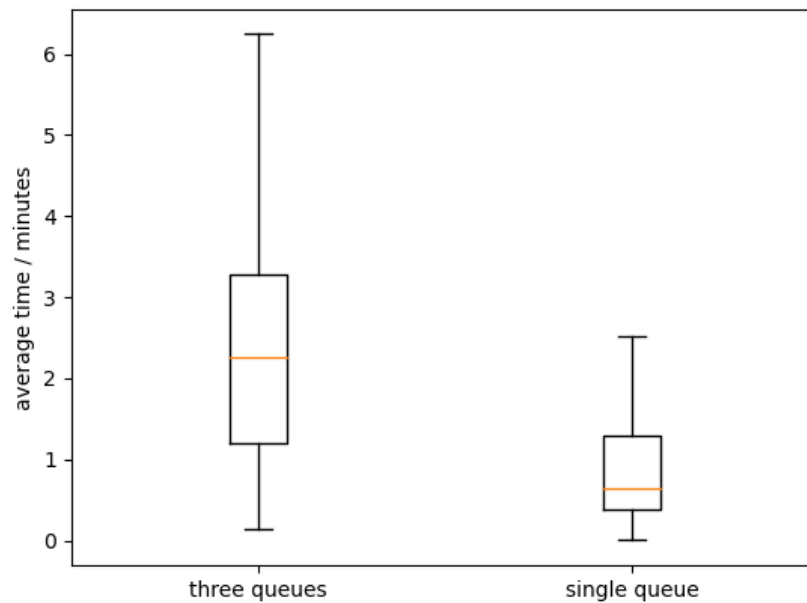
This three queue model, where each server specialises on a particular task, is compared with the model shown in the right pane of figure 1 where a single M/M/3 queue is displayed. In this model, a customer interacts with a single server who will both take their order and prepare their drink. The expected time  $S'$  taken to serve each customer in this model can be calculated as follows:

$$\mathbb{E}(S') = E(S) + 0.7\mathbb{E}(C) + 0.3\mathbb{E}(T) = 1 + 0.7 \times 5 + 0.3 \times 2 = 5.1 \quad \text{minutes}$$

Arrivals, meanwhile, are modelled in the same way they are modelled in the three-queue model.

### 3 Results

The efficiencies of the two models for running the cafe were compared by running simulations of a single day in the life of the cafe (10 hours) using queuing tool [1]. The average time customers spent queuing and being served for each of the two modes of operation was then estimated. In our simulation of a cafe where one server collects orders, one makes tea, and one makes coffee, 139 customers were served during the 10-hour day. These customers spent an average of 2.42 minutes queuing and getting their drinks. If we assume that the errors on

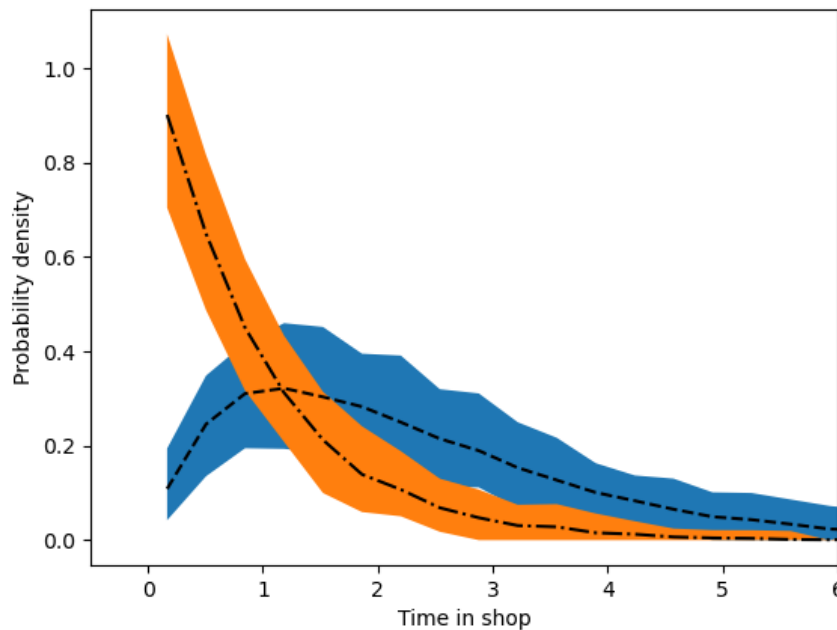


**Fig. 2.** Box plots showing the distribution of times people spend waiting in queues and being served. The distribution of times for the customers who enter the coffee shop during a single day is indicated.

this estimate are normally distributed, there is a 90% chance that the expected service time for this mode of operation is between 2.18 and 2.66 minutes.

In the simulation of a cafe where servers take orders and then prepare drinks for customers, 150 customers were served during the 10-hour day. These customers spent an average of 1.04 minutes queuing and getting their drinks. If we assume that the errors on this estimate are normally distributed, there is a 90% chance that the expected service time for this mode of operation is between 0.90 and 1.19 minutes. Thus, we can conclude that having the servers take customers' orders and then prepare their drinks is more efficient than having each server take specific roles in the cafe. Furthermore, this conclusion is reinforced when we visualise the distribution of times it took customers to get served. Figure 2 shows box plots of the service times of the customers who came into the cafe during our simulated days. The box plot on the left indicates the distribution of times for the model where one server is responsible for taking orders, one server is responsible for making teas, and one server is responsible for making coffees. You can see that the customer spends longer in the cafe on average when it is organised in this way. Furthermore, the distribution of waiting times is considerably wider when the cafe is organised this way. It can take customers up to six minutes to get served when each server takes a different role. When they all perform all tasks, the longest it takes an individual customer to get served is just over 2 minutes.

To further investigate the differences in the times customers spend in the shop under the two models, I constructed estimates of the probability density functions for the random



**Fig. 3.** Estimates of the probability density functions for the random variable that measures the amount of time a customer spends in the shop waiting to order and get their drink. The dashed line shows the histogram obtained for the model where each server has a different task. The dot-dashed line is the histogram obtained for the model where servers perform all three tasks. As discussed in the main text, the coloured regions give a confidence limit around these estimates.

variables that measure the time customers spend queuing and waiting for their drink to be prepared. The result of this investigation is shown in figure 3. The histograms in this figure were constructed by simulating 100 days of operation for coffee shops operating in each of the two ways described in section 2. Separate histograms were then constructed from these 200 sets of waiting times. For each of the two modes of operation, I thus had 100 estimates for the probability density function of the waiting time. The dashed line in figure 3 indicates the result from averaging over all the estimates for the distribution of waiting times when one server is responsible for taking orders, one is responsible for making coffee, and one is responsible for making tea. The dot-dashed line, meanwhile, is obtained by averaging over all estimates of the distribution of times customers spend in-store when a single server looks after their order. The blue and orange shaded areas around these lines indicate the 90% confidence limit around these estimates for the probability density. These confidence limits were obtained by sorting the samples of the probability density function that we got from our simulations.

The distributions in figure 3 support the conclusions that we arrived at by looking at the distribution of times (see figure 2) customers spend queuing and being served in a simulated day. Customers will spend less time queuing on average if a single server is responsible for taking each customer's order and preparing their drink. Any specialisation and division of

tasks between the various servers increases the amount of time that customers have to spend waiting.

## 4 Conclusion

Adam Smith argued in the wealth of nations [2] that if tasks are separated so that labourers specialise in doing only one thing, organisations become more efficient. Our results suggest that the opposite is the case in this coffee shop. Customers are served more quickly when servers take customer orders and prepare their drinks. If one server is responsible for taking orders, one server is responsible for making coffee, and one is responsible for making tea, there is a statistically significant increase in the time it takes to deal with a customer's needs.

Very simple models for the queues have been used in this report, and there is much that could be done to improve the modelling of the cafe.\* Even if that is done, however, the conclusion will likely be the same. The task that takes the longest is preparing a cup of coffee. It seems likely that queues for coffee would build up if only one person making coffees. Investigating whether a backlog builds up in this part of the queue undoubtedly merits further investigation.

## 5 References

1. *Tool: A Network simulator* 2016.
2. Smith, A. *The wealth of nations* (1776).

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\*You should never write something as vague as this first sentence in your reports. You should be specific about the weaknesses in your model. I have been vague because of the nature of the exercise I have set you, which involves identifying and addressing the shortcomings of my models.